COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE212 | Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	9-10-26	**DATE SUBMITTED:**	9-10-26
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ACTIVITY NO. 7.1 **Managing Files and Creating Roles in Ansible**

1. PROCEDURE OUTPUT

Task 1:

1.

```
  GNU nano 7.2                    default_site.html
<!DOCTYPE html>
<html lang="en">
<head>
    <metacharset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>My Web Server</title>
</head>
<body>

    <h1>Welcome to Masongsong's Web Server</h1>

    <p>This webpage was deployed using Ansible.</p>
    <p>This file was copied from the local repository to the remote server.</p>

    <h2>John Marvin B. Masongsong</h2>

</body>
</html>
```

2.

```
 GNU nano 7.2                          site.yml *
- hosts: web_servers
  become: true
  roles:
    - role: webserver
      tags: web

  tasks:
    - name: copy default html  file for site
      tags: apache, apache2, httpd
      copy:
        src: files/default_site.html
        dest: /var/www/html/index.html
        owner: root
        group: root
        mode: '0644'
```

3.

```
TASK [webserver : install apache and php for CentOS servers] ******************
skipping: [192.168.56.104]
ok: [192.168.56.102]

TASK [webserver : start httpd (CentOS)] ***************************************
skipping: [192.168.56.104]
ok: [192.168.56.102]

TASK [webserver : start apache2 (Ubuntu)] *************************************
skipping: [192.168.56.102]
ok: [192.168.56.104]

TASK [copy default html  file for site] ***************************************
ok: [192.168.56.104]
ok: [192.168.56.102]

PLAY [db_servers] *************************************************************

TASK [Gathering Facts] ********************************************************
ok: [192.168.56.104]
ok: [192.168.56.102]
```

- The task has been copied to Ubuntu and CentOS managed nodes.

4.

CentOS

```
masongsong@Masongsong:~/CPE212_Masongsong$ ssh masong@192.168.56.102
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Thu Sep 10 08:37:34 2026 from 192.168.56.101
[masong@localhost ~]$ cat /var/www/html/index.html
<!DOCTYPE html>
<html lang="en">
<head>
    <metacharset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>My Web Server</title>
</head>
<body>

    <h1>Welcome to Masongsong's Web Server</h1>

    <p>This webpage was deployed using Ansible.</p>
    <p>This file was copied from the local repository to the remote server.</p>

    <h2>John Marvin B. Masongsong</h2>

</body>
</html>
[masong@localhost ~]$
```

Ubuntu

```
masongsong@Masongsong:~/CPE212_Masongsong$ ssh masongsong@192.168.56.104
Last login: Thu Sep 10 00:37:33 2026 from 192.168.56.101
masongsong@Masongsong:~$ cat /var/www/html/index.html
<!DOCTYPE html>
<html lang="en">
<head>
    <metacharset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>My Web Server</title>
</head>
<body>

    <h1>Welcome to Masongsong's Web Server</h1>

    <p>This webpage was deployed using Ansible.</p>
    <p>This file was copied from the local repository to the remote server.</p>

    <h2>John Marvin B. Masongsong</h2>

</body>
</html>
masongsong@Masongsong:~$
```

Original

```
 GNU nano 7.2                    default_site.html
<!DOCTYPE html>
<html lang="en">
<head>
    <metacharset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>My Web Server</title>
</head>
<body>

    <h1>Welcome to Masongsong's Web Server</h1>

    <p>This webpage was deployed using Ansible.</p>
    <p>This file was copied from the local repository to the remote server.</p>

    <h2>John Marvin B. Masongsong</h2>

</body>
</html>
```

- The output is the exact same from the original because it was mainly copied to both the remote servers.

5.



- There is now a folder/directory in github named “files” and both the inventory.ini and site.yml files have been updated.

Task 2:

1.

```
  GNU nano 7.2                                site.yml *
      update_cache: yes
      cache_valid_time: 3600
    when: ansible_distribution == "Ubuntu"

- hosts: workstations
  become: true

  tasks:
    - name: install unzip
      package:
        name: unzip

    - name: install terraform
      unarchive:
      src: https://releases.hashicorp.com/terraform/0.12.28.terraform_0.12.28_l>
      dest: /usr/local/bin
      remote_src: yes
      mode: 0755
      owner: root
      group: root

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/ Go To Line
```

2.

```
[workstations]
192.168.56.102 ansible_user=masong
192.168.56.104
```

3.

```
TASK [install updates (Ubuntu)] ************************************************
skipping: [192.168.56.102]
ok: [192.168.56.104]

PLAY [workstations] ************************************************************

TASK [Gathering Facts] *********************************************************
ok: [192.168.56.104]
ok: [192.168.56.102]

TASK [install unzip] ***********************************************************
ok: [192.168.56.104]
ok: [192.168.56.102]

TASK [install terraform] *******************************************************
changed: [192.168.56.104]
changed: [192.168.56.102]
```

- The installation of unzip and terraform was completed in both remote servers.

4.

```
masongsong@Masongsong:~$ terraform --version
Terraform v0.12.28

Your version of Terraform is out of date! The latest version
is 1.16.2. You can update by downloading from https://www.terraform.io/downloads
.html
masongsong@Masongsong:~$
```

- Terraform has been installed in the remote ubuntu node, but it is out of date.

Task 3:
1.

```
  GNU nano 7.2                          site.yml *
---
- hosts: all
  become: true
  pre_tasks:

  - name: update repository index (CentOS)
    tags: always
    dnf:
      update_cache: yes
    changed_when: false
    when: ansible_distribution == "CentOS"

  - name: install updates (Ubuntu)
    tags: always
    apt:
      update_cahce: yes
    changed_when: false
    when: ansible_distribution == "Ubuntu"
```

```
- hosts: all
  become: true
  roles:
    - base

- hosts: workstations
  become: true
  roles:
    - workstations

- hosts: web_servers
  become: true
  roles:
    - web_servers
```

```
- hosts: db_servers
  become: true
  roles:
    - db_servers

- hosts: file_servers
  become: true
  roles:
    - file_servers
```

2.

```
masongsong@Masongsong:~/CPE212_Masongsong/roles/workstations/tasks$ cd ~/CPE212_
Masongsong/roles
masongsong@Masongsong:~/CPE212_Masongsong/roles$ ls
base  db_servers  file_servers  web_servers  workstations
masongsong@Masongsong:~/CPE212_Masongsong/roles$ cd base
masongsong@Masongsong:~/CPE212_Masongsong/roles/base$ ls
tasks
masongsong@Masongsong:~/CPE212_Masongsong/roles/base$ cd -
/home/masongsong/CPE212_Masongsong/roles
masongsong@Masongsong:~/CPE212_Masongsong/roles$ cd db_servers
masongsong@Masongsong:~/CPE212_Masongsong/roles/db_servers$ ls
tasks
masongsong@Masongsong:~/CPE212_Masongsong/roles/db_servers$ cd -
/home/masongsong/CPE212_Masongsong/roles
masongsong@Masongsong:~/CPE212_Masongsong/roles$ cd file_servers
masongsong@Masongsong:~/CPE212_Masongsong/roles/file_servers$ ls
tasks
masongsong@Masongsong:~/CPE212_Masongsong/roles/file_servers$ cd -
/home/masongsong/CPE212_Masongsong/roles
masongsong@Masongsong:~/CPE212_Masongsong/roles$ cd web_servers
masongsong@Masongsong:~/CPE212_Masongsong/roles/web_servers$ ls
tasks
masongsong@Masongsong:~/CPE212_Masongsong/roles/web_servers$ cd -
/home/masongsong/CPE212_Masongsong/roles
masongsong@Masongsong:~/CPE212_Masongsong/roles$ cd workstations
masongsong@Masongsong:~/CPE212_Masongsong/roles/workstations$ ls
tasks
masongsong@Masongsong:~/CPE212_Masongsong/roles/workstations$ cd -
/home/masongsong/CPE212_Masongsong/roles
masongsong@Masongsong:~/CPE212_Masongsong/roles$
```

3.

Directory	Code
base	GNU nano 7.2 main.yml * --- - name: update repository index (CentOS) tags: always dnf: update_cache: yes changed_when: false when: ansible_distribution == "CentOS" - name: install updates (Ubuntu) tags: always apt: update_cache: yes changed_when: false when: ansible_distribution == "Ubuntu"
db_servers	GNU nano 7.2 main.yml * --- - name: install mariadb package (CentOS) tags: centos, db, mariadb dnf: name: mariadb-server state: latest when: ansible_distribution == "CentOS" - name: install mariadb package (Ubuntu) tags: db, mariadb, ubuntu apt: name: mariadb-server state: latest when: ansible_distribution == "Ubuntu" - name: "Mariadb - Restarting/Enabling" tags: db, mariadb service: name: mariadb state: restarted enabled: true

file_servers

```
 GNU nano 7.2                                  main.yml
---
- name: install samba package
  tags: samba, file
  package:
    name: samba
    state: latest

- name: start and enable samba
  tags: samba, file
  service:
    name: smb
    state: started
    enabled: true
```

web_servers

```
 GNU nano 7.2                                  main.yml *
---
- name: install apache and php for Ubuntu servers
  tags: apache, apache2, ubuntu
  apt:
    name:
      - apache2
      - libapache2-mod-php
    state: latest
    update_cache: yes
  when: ansible_distribution == "Ubuntu"

- name: install apache and php for CentOS servers
  tags: apache, centos, httpd
  dnf:
    name:
      - httpd
      - php
    state: latest
  when: ansible_distribution == "CentOS"

- name: start httpd (CentOS)
  tags: apache, centos, httpd
  service:
    name: httpd
    state: started
    enabled: true
  when: ansible_distribution == "CentOS"

- name: start apache2 (Ubuntu)
  tags: apache, apache2, ubuntu
  service:
    name: apache2
    state: started
    enabled: true
  when: ansible_distribution == "Ubuntu"
```

workstations	GNU nano 7.2 main.yml --- - name: install unzip package: name: unzip - name: install terraform unarchive: src: https://releases.hashicorp.com/terraform/0.12.28 dest: /usr/local/bin remote_src: yes mode: 0755 owner: root group: root

4.

```
TASK [Gathering Facts] ***********************************************
ok: [192.168.56.102]
ok: [192.168.56.104]

TASK [base : update repository index (CentOS)] ***********************
skipping: [192.168.56.104]
ok: [192.168.56.102]

TASK [base : install updates (Ubuntu)] *******************************
skipping: [192.168.56.102]
ok: [192.168.56.104]
```

- Base roles are working

```
PLAY [workstations] **************************

TASK [Gathering Facts] ***********************
ok: [192.168.56.104]
ok: [192.168.56.102]

TASK [workstations : install unzip] **********
ok: [192.168.56.104]
ok: [192.168.56.102]

TASK [workstations : install terraform] ******
ok: [192.168.56.104]
ok: [192.168.56.102]
```

- Workstation roles have downloaded unzip and terraform

```
PLAY [web_servers] ************************************************************

TASK [Gathering Facts] ********************************************************
ok: [192.168.56.102]
ok: [192.168.56.104]

TASK [web_servers : install apache and php for Ubuntu servers] *****************
skipping: [192.168.56.102]
ok: [192.168.56.104]

TASK [web_servers : install apache and php for CentOS servers] *****************
skipping: [192.168.56.104]
ok: [192.168.56.102]

TASK [web_servers : start httpd (CentOS)] *************************************
skipping: [192.168.56.104]
ok: [192.168.56.102]

TASK [web_servers : start apache2 (Ubuntu)] ***********************************
skipping: [192.168.56.102]
ok: [192.168.56.104]

TASK [web_servers : copy default html file for site] ***************************
ok: [192.168.56.104]
ok: [192.168.56.102]
```

- Web_servers have installed and started apache and php for both Ubuntu and centos, as well as copied the default html file for site.

```
PLAY [db_servers] ***************************************************

TASK [Gathering Facts] **********************************************
ok: [192.168.56.104]
ok: [192.168.56.102]

TASK [db_servers : install mariadb package (CentOS)] ***************
skipping: [192.168.56.104]
ok: [192.168.56.102]

TASK [db_servers : install mariadb package (Ubuntu)] ***************
skipping: [192.168.56.102]
ok: [192.168.56.104]

TASK [db_servers : Mariadb - Restarting/Enabling] ******************
changed: [192.168.56.102]
changed: [192.168.56.104]
```

- Db_servers have installed and enabled the mariadb package on both Centos and Ubuntu.

```
PLAY [file_servers] *************************************************

TASK [Gathering Facts] **********************************************
ok: [192.168.56.104]
ok: [192.168.56.102]

TASK [file_servers : install samba package] *************************
ok: [192.168.56.102]
changed: [192.168.56.104]

TASK [file_servers : start and enable samba] ************************
ok: [192.168.56.104]
ok: [192.168.56.102]
```

- File_servers have installed and enabled the samba package in both Ubuntu and Centos.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Answer the following:

1. What is the importance of creating roles?

- **Creating roles allows you to manage and reuse different files that are often clumped up in your workstation, it makes it easier to determine which role does what and besides that it contains only the tasks you want to do for that specific role.**

2. What is the importance of managing files?

- **It reduces clutter from your workstation making it more visibly pleasing and less confusing when trying to navigate through your file system.**

Conclusion:

I have learned a lot from this HOA mainly how to manage files and create roles. Roles allow you to pinpoint which tasks are used for the job while managing files allow you to organize your workstation allowing for a more visibly pleasing experience when navigating your file system. It was quite hard at first when trying to create the roles itself and also segmenting the old site.yml to its own meta group, but after getting past that I understood most of it and was able to run the final output without fail. All in all I had fun in understanding what and how managing files and creating roles does to your workstation, and I hope to use this information in my future HOAs.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **NA**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **NA**

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